

WIFI Security & Standards

WPA2 & WPA3

WPA = WiFi Protected Access

- * It works to authenticate devices and protect wireless communication from unauthorized access.

WEP → WPA → WPA2 → WPA3

1) WEP = Wired Equivalent Privacy

- * old WiFi Security method

- * In 1997 introduced

- * RC4 Encryption used, Cracked Easily

2) WPA: = WiFi Protected Access

- * Introduced 2003

- * Cracks are temporary replaced with WEP

- * uses TKIP [Temporal Key Integrity Protocol]

WPA 2

* Introduced in 2004

* uses AES - CCMP Encryption

* WPA2 personal Commonly uses a PSK

[Pre-shared key]

the WIFI password is used as shared secret for

Security process * Unauthorized access stops

* Still widely Supported * Data sent over WiFi

* uses PSK + 4 way handshake for Encryption keys

WPA3

* Introduced in 2018

* uses SAE [Simultaneous Authentication of Equals]
for WPA3 personal

* Its has strong protection against password
guessing attacks / brute force and stronger

SSID = Name of WIFI

Service Set Identifier

ESP32 (Temp 32)

Encryption

↓
Router

↓
Decryption

↓

Temp 32

WIFI - STA [Station]

It means the Esp32 operates as a WIFI Client

Home Router

WIFI Access point



WIFI



ESP 32 [STA]

Code

WIFI.mode (WIFI_STA)

STA vs AP

STA = Connects to another WIFI N/W's

AP = Creates It's own WIFI N/W

We can use both AP + STA mode and

We can use WPA2 and WPA3 too

* STA is for Operating and Interfacing

" WPA2 / WPA3 protects the WIFI link by authenticating the ESP32 (or) any device and Establishing Encryption keys, so unauthorized devices can't join the protected N/W & can't capture wireless traffic can't be directly read

802.11 Standard

It is the family of Standard underlying WiFi

* It Specifies how WiFi devices Communicate over Wireless W/L's

WiFi Router/
Access point → 802.11 → W/L radio waves → phone/
laptop

① Router Creates a WiFi network [SSID, PSK]



Your device See the WiFi



phone and router Communicate using 802.11

phone Router



Data is Converted into radio Signals



device Rxes the data

802.11 how different versions are

802.11

- 802.11b → 11 mbps → 2.4 GHz
- 802.11g → 54 mbps → 2.4 GHz
- 802.11n → 600 mbps → MIMO
- 802.11ac → 6.9 Gb → 5 GHz → MU-MIMO
- 802.11ax → 9.6 Gb → WiFi 6 → OFDMA

MIMO = multiple Input Multiple outputs

802.11 ax [WiFi 6]

One common

- OFDMA
- Improved MU-MIMO (multi users mimo)
- 1024 - QAM [2^{10} Quadrature Amp mod]
- BSS Coloring [Basic Service Set]
- TWT [Target Wake Time]

WiFi - 802.11 IEEE

WiFi - AUTH -

WPA3 - PSK

Ethernet - IEEE 802.3

WPA2 & WPA3

BLE / BE → BE Specification

Zigbee → IEEE 802.15.4 based

WPA2 - Authentication process

① Device searches for WIFI

- device scans for near by WIFI networks
- It finds required SSID

② device requests Connection to WIFI

- I want to connect to WIFI
- Router checks the Security, it's configured with WPA2 - personal
- The device and router use the WIFI password as Secret

③ password is converted into Security key

- password + SSID = PMK (Pairwise master key)
- The password itself is not sent directly to the router

④ 4-way handshake starts

- device and router exchange some random values
- These are called Nonces
- Router sends nonce, device generates nonce

Both Sides Create another key

- using the pmk + random values + device / router addresses
- both Calculate PTK
- PTK = pairwise Transient key

Both sides Verify Each other

They use MIC to check that the handshake msg are valid and haven't been modified

→ once handshake Completed, device & router have the same Encryption keys

Data is Encrypted : WPA2 uses AES-CCMP to protect data

Now Comm. Starts b/w router & device

"

MIC = msg Integrity Code, It used to verify the integrity/authenticity of handshake msg

req Encryption & keys : After authentication,

device & router needs Cryptographic keys to protect the actual WIFI data, PTK is derived by both sides independently, calculate both PTK and same PTK will make them done with